



Ques. WHAT IS AN IoT PLATFORM? NAME SOME FREE IoT PLATFORMS SUITABLE FOR POPULAR DEVELOPMENT BOARDS LIKE ARDUINO OR RASPBERRY PI?

Pamarthi Kanakaraja

Ans. An IoT (Internet of Things) platform is an essential part of a large IoT ecosystem that supports and connects all components, relays and information using transmission control protocol/Internet protocol (TCP/IP) within the system. In other words, an IoT platform connects hardware to the cloud, speeds up development of applications for connected devices, and takes care of scalability and cross-device compatibility.

There are many different IoT platforms available from different vendors with unique solutions for different customers and businesses. Some IoT platforms available for Arduino and Raspberry Pi are listed below, along with their websites.

- Cayenne: <https://cayenne.mydevices.com/>
- Blynk: www.blynk.cc/
- Thingspeak: <https://thingspeak.com/>
- Thingsboard.io: <https://thingsboard.io/>
- ubidots: <https://ubidots.com/>
- opendevise: <http://opendevise.io/>

Q2. WHAT ARE THE DIFFERENCES BETWEEN COPYRIGHTS, PATENTS AND TRADEMARKS? WHY ARE THESE IMPORTANT FOR STARTUPS OR ENTREPRENEURS?

A. Samiuddhin

A2. Intellectual property (IP) rights include copyrights, patents, trademarks, industrial design rights and trade secrets, among others. There are differences between these depending on the work one is trying to protect. He or she may use one or more IP tools to protect the work.

Copyright. This protects original

textual works and visual or artistic expressions. It covers a wide range of creative, intellectual or artistic forms including poems and other literary works, photographs, motion pictures, choreography, musical compositions, paintings, drawings, sculptures, computer software, radio and television broadcasts, and industrial designs.

To qualify for copyright, a work must meet minimal standards of originality. The copyright expires after a set period of time—the life of the author plus either 50 or 70 years.

The copyright law requires the use of a copyright notice, consisting of the copyright symbol (©), the Copr. abbreviation or the word Copyright, followed by the year of the first publication of the work and the name of the copyright holder. The copyright notice for sound recordings or other audio works is denoted by copyright symbol ©, where P indicates phonorecord. Also, the phrase “All rights reserved” is used to assert copyright.

Normally, copyright is automatic—official registration with any government office is not required. However, in jurisdictions where registration is required, it serves as evidence of a valid copyright, which makes it easy for the copyright holder to seek statutory damages.

In India, Copyright Act 1957 governs the subject of copyright law.

Patents. A patent gives the owner the right to exclude others from making, using, selling and importing an invention for a limited period of time, usually 20 years. An invention can be a device, structure, process, machinery and so on. The grant and enforcement of patents are governed by national laws, and by international treaties.

In India, under the provisions of Section 159 of Patents Act, 1970, the Central government has been empowered to make rules for implementing the Act and regulating patent administration. Since then, Indian Patent Rules were amended many a times, the latest being in 2018.

You can get more information from

the website <http://www.ipindia.nic.in/history-of-indian-patent-system.htm>

Trademarks. A trademark could be a name, word, phrase, logo, symbol, design, image or expression that identifies products or services of a particular source from those of others or a combination of these elements. A trademark owner can be an individual, business organisation or any legal entity. A typical example of a trademark is shown in the figure below. A trademark may be located on a package, label, voucher or the product itself.



A typical example of trademark

Trademarks give the owner exclusive right over certain images and phrases, and the right to prevent others from using a similar mark. These may be designated by the following symbols:

- Letters TM in superscript for an unregistered trademark to promote or brand goods.
- Letters SM in superscript for an unregistered service mark to promote or brand services.
- Letter R surrounded by a circle for a registered trademark.

You can visit https://en.wikipedia.org/wiki/Indian_trademark_law for more information.

Having a clear understanding of IP is a crucial part of building a new business venture. There is always a competitive advantage and potential value to investors when there is clarity in the IP startup portfolio. Therefore a startup or entrepreneur should at least have some basic knowledge on IP protection. This knowledge can give an advantage in handling risks arising from IP-related issues, managing opportunities and earning profits from company's assets.

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender.